

Wiener Filtering for Speech Enhancement

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Problem Formulation

Objective: Suppress background noise from a speech recording.

Signal Model

The observed noisy signal $y(n)$ is modeled as:

$$y(n) = x(n) + v(n) \quad (1)$$

where:

- $x(n)$: Clean speech (unknown)
- $v(n)$: Background noise (aircraft engine/cabin noise)

Goal: Design a linear estimator to minimize the mean-square error (MSE):

$$J = \mathbb{E} [(x(n) - \hat{x}(n))^2] .$$

Methodology: Noise/Speech Segment Selection

Challenge: Estimating noise statistics from a single noisy recording.

Strategy: Short-Time Energy Selection

- ① Seed: First 0.7 s assumed noise-dominant.
- ② Framing: 20 ms frames with 10 ms hop.
- ③ Energy percentiles:
 - Lowest 15% energy → Noise set
 - Highest 85% energy → Speech set

Methodology: AR Modeling

Both speech and noise are modeled as Auto-Regressive (AR) processes to obtain smooth PSD estimates.

AR Model Definition

Speech: $A_x(q^{-1})x[n] = e_x[n]$, Noise: $A_v(q^{-1})v[n] = e_v[n]$, where $A(q^{-1}) = 1 + \sum_{k=1}^p a_k q^{-k}$.

$$\Phi_{xx}(e^{j\omega}) = \frac{\sigma_x^2}{|A_x(e^{j\omega})|^2}, \quad \Phi_{vv}(e^{j\omega}) = \frac{\sigma_v^2}{|A_v(e^{j\omega})|^2}.$$

Parameter estimation:

- Implemented batch least squares (normal equations).
- Produces A_x , A_v and innovation variances σ_x^2 , σ_v^2 .

Assumptions and Motivated Approximations

Key Assumptions

- Additive model: $y(n) = x(n) + v(n)$.
- Uncorrelated components: $x(n)$ and $v(n)$ are assumed uncorrelated.
- Silence heuristic: low short-time energy frames are noise-dominant.
- AR suitability: $x(n)$ and $v(n)$ are well-approximated by AR models for spectral estimation.

Motivated Choices / Approximations

- Finite-length implementations: non-causal and causal Wiener filters are implemented as FIR filters via truncation ($L_{nc} = L_c = 1025$).
- Model/complexity trade-off: $p_X = p_V = 14$, $M = 192$ chosen to balance spectral smoothness and computational cost.
- Noise scaling: $\beta = 20$ selected based on listening tests to increase suppression of broadband aircraft noise.

Three different Wiener filters were implemented and compared:

1. Ideal Non-Causal

$$H_{nc}(e^{j\omega}) = \frac{\Phi_{xx}}{\Phi_{xx} + \beta\Phi_{vv}}$$

Theoretical benchmark (offline).

2. FIR Wiener (Wiener-Hopf)

$$\mathbf{R}_{yy}\mathbf{h} = \mathbf{r}_{yx}$$

Real-time feasible, finite length
($M = 192$).

3. Causal Wiener (Spectral Factorization)

$$H_c = \frac{1}{\Phi_{yy}^+} \left\{ \frac{\Phi_{xx}}{\Phi_{yy}^-} \right\}_+$$

IIR in theory; implemented via FIR
truncation ($L_c = 1025$).

Implementation Parameters

Parameter	Value
Noise scaling (β)	20.0
AR order (p_X, p_V)	14
FIR Wiener length (M)	192
FFT size (N_{fft})	8192
Truncation (L_{nc}, L_c)	1025
Noise seed	0.7 s
Frame / hop	20 ms / 10 ms
Percentiles	15% / 85%

Trade-off ($\beta = 20$):

- Aircraft noise is broadband and overlaps with speech bands.
- Standard $\beta = 1$ yielded insufficient suppression.
- Increased β to 20 for stronger attenuation.
- Side effect is speech distortion.

Results: Frequency Response

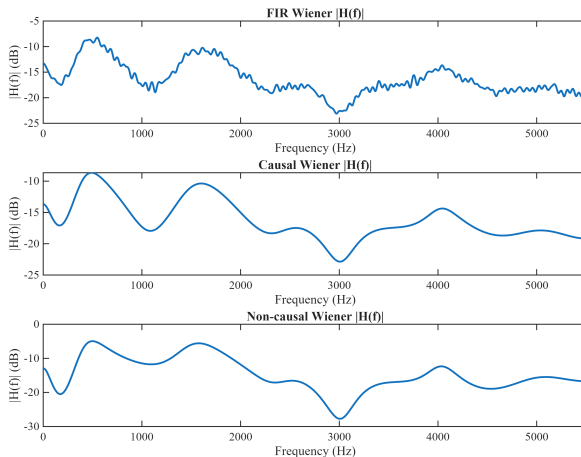


Figure 1: Magnitude response $|H(f)|$ for the three Wiener designs ($\beta = 20$).

Results: Time Domain Waveforms

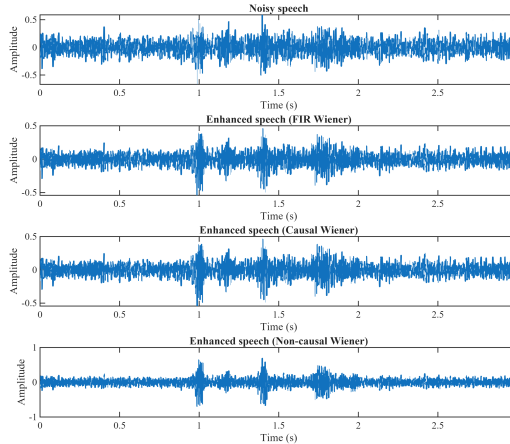


Figure 2: Time-domain comparison: noisy input and enhanced outputs (first few seconds).

- Non-causal provided the strongest noise-floor reduction; causal and FIR yielded milder suppression.
- Short-time energy percentiles exploited silent periods to obtain more reliable noise/speech samples.
- Strong suppression with $\beta = 20$ improved noise reduction but introduced audible distortion, reflecting the inherent estimation trade-off.

Thank you!